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Studierichting: Informatica
Oefeningenreeks 5B nr. 1.5

$$\begin{aligned} & \int_0^{+\infty} \frac{dx}{1+x^2} \\ &= \lim_{b \rightarrow +\infty} \int_0^b \frac{dx}{1+x^2} \\ &= \lim_{b \rightarrow +\infty} [\text{Bgtan } x]_0^b \\ &= \lim_{b \rightarrow +\infty} \text{Bgtan } b - \text{Bgtan } 0 \\ &= \lim_{b \rightarrow +\infty} \text{Bgtan } b = \frac{\pi}{2} \end{aligned}$$

References